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Lanthanum nitride LaN_9 featuring azide units: the first metal nine-nitride as a high-energy-density material†

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High-pressure phase diagrams of the La–N binary system were systematically constructed using the CALYPSO method and first-principles calculations. In addition to the pressure-induced La–N compounds reported previously, we have uncovered a hitherto unknown LaN_9 structure in $Pm\bar{3}$ symmetry stabilized within a narrow pressure range of 20–24.5 GPa. Notably, LaN_9 stands as the first thermodynamically stable metal nine-nitrogen compound, featuring centrosymmetric linear N_3 anion units and an edge-sharing LaN_{12} icosahedron. Charge transfer between the La and N atoms plays a crucial role in facilitating structural stability. Furthermore, we identified a novel Cm phase for LaN_8 , which has a lower enthalpy compared to the previously reported phase. N atoms in Cm LaN_8 are polymerized into infinite N_∞ chains. Calculations demonstrate the potential recoverability of LaN_9 and Cm LaN_8 under atmospheric conditions while preserving their initial polynitrogen configuration. From the perspective of detonation pressure and detonation velocity, LaN_9 and Cm LaN_8 exhibit excellent explosive performance in comparison to TNT and HMX, with estimated energy densities of 0.9 and 1.54 kJ g^{-1} , respectively, indicating their potential utility as high-energy-density materials.

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1. Introduction

The design and synthesis of environment-friendly high-energy-density materials (HEDMs) have been a hot topic within the realms of condensed matter physics and materials science.^{1,2} Among the diverse spectrum of HEDMs, polynitrogen compounds have garnered substantial attention owing to their high energy density, and the environmentally benign byproduct.^{3,4} The substantial energy density exhibited by polynitrogen compounds can be attributed to the significant difference in average bond energies between N–N single/double bonds and triple bonds.⁵ To date, several pure polynitrogen compounds have been predicted theoretically and a number of them have been synthesized experimentally under high-pressure high-temperature conditions (e.g. cg-N ,⁶ Pba2-N ,⁷ $\text{P4}_2bc\text{-N}$,⁸ black phosphorus-structured nitrogen⁹). However, their practical application as HEDMs is nearly unattainable due to their unquenchability under atmospheric conditions.

Metal atoms play a key role in stabilizing nitrogen-bearing polyanions through charge transfer mechanisms at high pressures. Extensive theoretical and experimental investigations have demonstrated that the majority of metals spanning the periodic table, including alkali metals,^{10–19} alkaline-earth metals,^{20–24} transition metals,^{25–30} and rare-earth metals,^{31–34} are capable of engaging in reactions with nitrogen atoms at high pressure (please see the ESI†). This capability facilitates the formation of a diverse range of polynitrogen frameworks. Of particular interest are frameworks encompassing a wide array of polynitrogen units, including azides (N_3^-) as exemplified in alkali metal azides AN_3 ($M = \text{Li, Na, K, Rb, or Cs}$),^{10,11,18,35–37} pentazolate (N_5^-) as observed in MN_5 ($M = \text{Li, Na, K, Rb, Cs, Ca, Sr, Ba, Cu, Ta}$)^{15,18,19,22–24,38–40} or MN_{10} ($M = \text{Be, Mg, Ba}$),^{24,41,42} benzene (N_6^-) as manifested in MN_6 ($M = \text{W, Sn}$),^{27,43} and N_∞ anion chains as demonstrated in GdN_6 , ReN_8 , HfN_{10} , and FeN_6 .^{25,34,44,45} To the best of our knowledge, metal nitrides characterized by a nitrogen-to-metal ratio of 9:1 have not yet been discovered.

Lanthanum (La) shares a similar electronegativity to alkali and alkaline-earth metals and has a significant number of outer-shell electrons, rendering it well-suited for the formation of nitrogen-rich nitrides. This is supported by the theoretical proposal of LaN_8 with a Cc space group when subjected to high-pressure conditions.⁴⁶ In this study, we present our investigation of the high-pressure phase diagram of La–N, with

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particular emphasis on nitrogen-rich species, employing CALYPSO crystal structure searches in conjunction with first-principles calculations. Within a narrow pressure range of 20–24.5 GPa, we have uncovered a previously undiscovered $Pm\bar{3}$ LaN_9 which incorporates azide anion units and a LaN_{12} icosahedron. Importantly, it is the first known thermodynamically stable metal compound containing nine nitrogen atoms. Additionally, our predictions indicate the existence of a novel Cm phase of LaN_8 with a lower enthalpy in comparison to the previously reported Cc phase. Cm LaN_8 is characterized by infinite N_∞ chains. The quenchable recoverability of $Pm\bar{3}$ LaN_9 and Cm LaN_8 under atmospheric conditions has been examined using phonon spectra and *ab initio* molecular dynamics simulations. Our energy density, detonation velocity, and pressure calculations reveal the potential application of LaN_9 and Cm LaN_8 as HEDMs.

2. Computational methods

We conducted systematic structure searches for LaN_x ($1 \leq x \leq 10$) systems within simulation cells containing 1 to 4 formula units in the pressure range of 0–100 GPa (0, 20, 25, 50, and 100 GPa) using the CALYPSO method.^{47–50} Local structural relaxations and electronic property calculations were carried out using density functional theory⁵¹ within the generalized gradient approximation,⁵² as implemented in the Vienna *ab initio* simulation package.⁵³ We employed the projector augmented-wave approximation,⁵⁴ treating $2s^22p^3$ and $5d^16s^2$ as valence electrons for N and La, respectively. Plane-wave cutoff energy of 550 eV and a dense Monkhorst–Pack grid⁵⁵ spacing of $2\pi \times 0.03 \text{ \AA}^{-1}$ were employed. Convergence criteria for energy and forces were set at 10^{-5} eV and $0.001 \text{ eV \AA}^{-1}$, respectively. To assess the dynamic stability of the predicted structures, phonon spectra calculations were conducted using the direct supercell methods within the PHONOPY code.^{56,57} Additionally, we performed *ab initio* molecular dynamics (AIMD) simulations to investigate the thermal stability of the predicted structures, utilizing the Nosé–Hoover chain

thermostat⁵⁸ with a total simulation time of 20 ps. We calculated the detonation velocity and detonation pressure using the Kamlet–Jacobs semiempirical equation.⁵⁹

3. Results and discussion

3.1 Phase stability of the La–N system under pressure

We conducted a comprehensive investigation into the stability of the La–N system by calculating formation enthalpies relative to the constituent elements and constructed convex hulls at various pressures (0, 20, 25, 50, and 100 GPa), as depicted in Fig. 1(a). At ambient pressure, only LaN and LaN_2 are found to be thermodynamically stable. However, as pressure increases, four stoichiometries (LaN_4 , LaN_9 , LaN_8 , and LaN_3) become stable against decomposition into their elemental constituents at several pressure points. We have calculated the enthalpy difference of the four stable stoichiometries to elucidate the pressure conditions under which these stoichiometries remain stable, as shown in Fig. 1(b). LaN_4 is energetically stable over a large pressure range of 14–83 GPa and undergoes a pressure-induced structural transition from the monoclinic $C2/c$ to triclinic $P\bar{1}$ structure at 39 GPa. Cm LaN_8 and $C2/c$ LaN_3 exhibit stability above 24 and 48 GPa, respectively, throughout the entire pressure range up to 100 GPa, which was considered in our analysis. In contrast, cubic LaN_9 with $Pm\bar{3}$ symmetry is stable within a very narrow pressure window, specifically between 20 and 24.5 GPa. Fig. S1 (ESI†) depicts this stable pressure range of $Pm\bar{3}$ LaN_9 by thoroughly analyzing all potential decomposition pathways. Additional information on the crystal structures can be found in Table SI (ESI†). Taking into consideration the recoverability of LaN_9 and LaN_8 at ambient pressure, our attention is primarily directed towards these two materials.

3.2 The first metal nine-nitrogen compound, LaN_9 , featuring azide units

Similar to alkali metal azides, LaN_9 is an ionic solid, *i.e.*, $\text{La}^{3+}(\text{N}_3)_3^-$ at high pressure. It crystallizes in a cubic Bravais

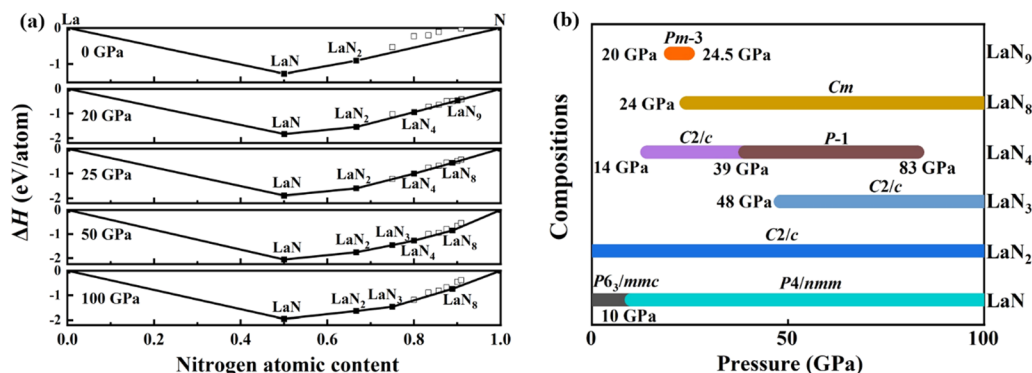


Fig. 1 (a) Convex hull diagrams of the La–N system at high pressures. The stable and metastable phases are marked by solid and hollow blocks, respectively. (b) Phase diagrams of the predicted stable La–N compounds in the pressure range of 0–100 GPa. The structures of α -N (0–0.23 GPa),⁶⁰ γ -N (0.23–9.5 GPa),⁶¹ and ϵ -N (9.5–110 GPa)⁶² for nitrogen, as well as dhcp (0–8 GPa), dfcc (8–59 GPa), and fcc (60–100 GPa)^{63–65} for La, are considered as reference structures in the calculations of the formation enthalpies.

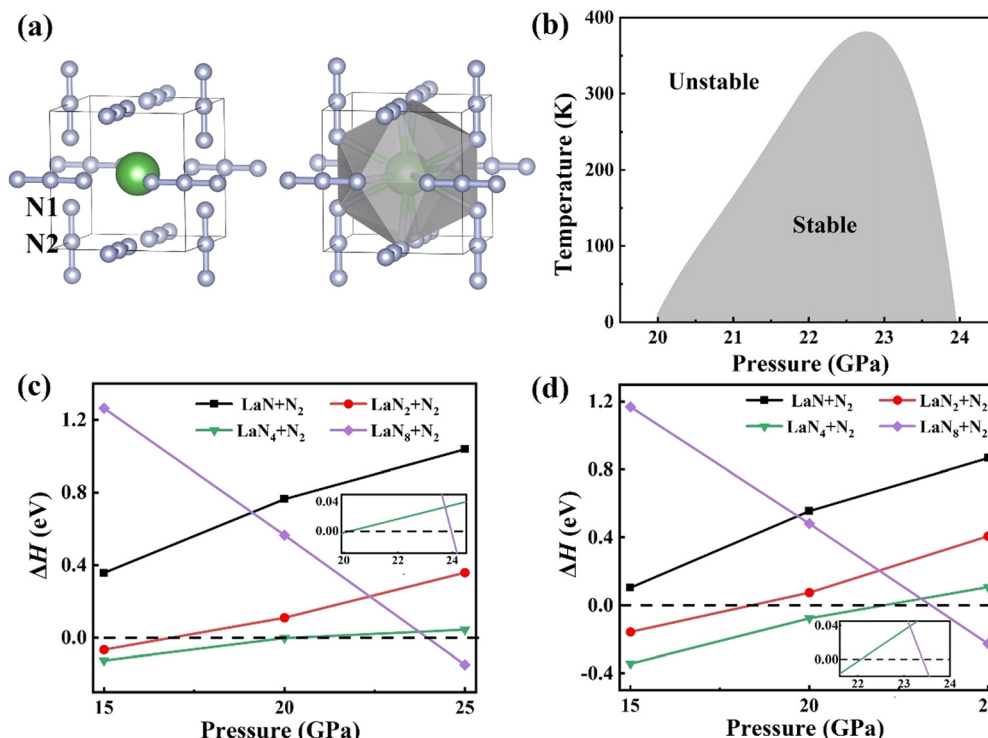


Fig. 2 (a) Crystal structure of LaN_9 at 20 GPa. The light gray and green spheres represent N atoms and La atoms, respectively. (b) Pressure–temperature phase diagram of LaN_9 determined by Gibbs free energy. The formation enthalpy of LaN_9 (c) with zero-point energy and (d) at 300 K as a function of pressure.

lattice, space group $Pm\bar{3}$, with one formula unit per cell. In LaN_9 , body-centered La cations and centrosymmetric linear N_3 anions are both packed in the NaCl-type structure. Of particular note, the terminating nitrogens (N1) of the azide anions are coordinatively bonded with two La cations, and each La cation is further coordinated by the 12 nearest terminal nitrogens of the azide anions. This coordination results in the formation of edge-sharing LaN_{12} icosahedra, as illustrated in Fig. 2(a). The azide units within LaN_9 are notably well separated. The length of the N–N bond is 1.17 Å, similar to that in alkali metal azides.

We have calculated the Gibbs free energy *versus* temperature by considering the vibrational contributions through the quasi-harmonic approximation and constructed a pressure–temperature phase diagram of LaN_9 . As illustrated in Fig. 2(b), elevated temperatures diminish the stable pressure range of LaN_9 . Specifically, at 0 K, the stable pressure range spans from 20 to 24 GPa, as shown in Fig. 2(c), whereas at 300 K, the range contracts to 22–23.4 GPa, as shown in Fig. 2(d). It is noteworthy that the synthesis of LaN_9 necessitates a pressure of 20 GPa, a requirement found to be comparatively lower than that of other metal nitrides. For instance, LiN_5 ,¹² CsN_5 ,⁴⁰ MgN_4 ,²¹ and WN_6 ²⁷ require 45, 60, 50, and 126 GPa, respectively. Consequently, LaN_9 would be formed by cold compressing suitable precursors.

To assess the dynamical and thermal stability of LaN_9 under high-pressure conditions, we conducted phonon dispersion curve calculations and AIMD simulations at 20 GPa. As depicted in Fig. 3(a), our analysis reveals that there are no

imaginary frequencies within the first Brillouin zone, thus confirming its dynamical stability. Furthermore, MD simulations at 300 K for 20 ps indicate that the structure of LaN_9 does not exhibit any evident structural degradation (Fig. 3(b)), underscoring its thermal stability. Phonon spectrum calculation and AIMD simulation were conducted under atmospheric conditions, as shown in Fig. 3(c) and (d), confirming the recoverability of LaN_9 under atmospheric conditions. Additionally, we have demonstrated that the elastic constants of LaN_9 , as detailed in Table SII (ESI†), meet the mechanical stability criteria for the cubic phase.⁶⁶ Considering the metastability of LaN_9 at ambient pressure, insights from Sun *et al.*'s analysis⁶⁷ on metastability in inorganic crystals prove valuable. The concept of “remnant metastability” suggests that phases stable under specific thermodynamic conditions may retain their metastable state when those thermodynamic constraints are removed. Applying this concept to LaN_9 , it is reasonable to assume that once synthesized at high pressure, LaN_9 will remain stable even at ambient pressure.

To evaluate the overlap strength of the N–N and La–N bonds in LaN_9 , we analyzed the projected crystal orbital Hamilton population (pCOHP) and integral COHP (ICOHP), as depicted in Fig. 4(a). The negative and positive pCOHP values denote bonding and antibonding interactions, respectively. The integrated COHP (ICOHP) serves as an indicator of bonding strength. It is evident that both the La–N and N–N bonding states are fully occupied, while the antibonding states are only

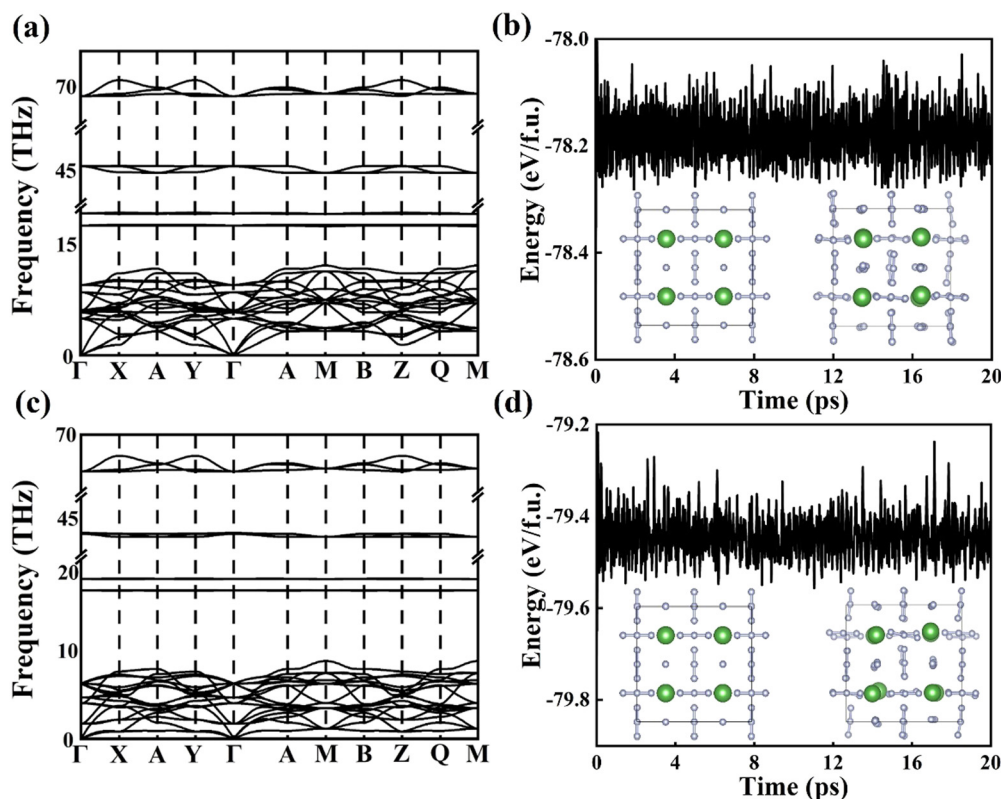


Fig. 3 (a) Phonon dispersion curves and (b) energy fluctuations during MD simulations (300 K) of LaN_9 at 20 GPa. (c) Phonon dispersion curves and (d) energy fluctuations during MD simulations (300 K) of LaN_9 at 0 GPa. The insets in (b) and (d) are the structural snapshots at the first and end of the 20-ps simulations.

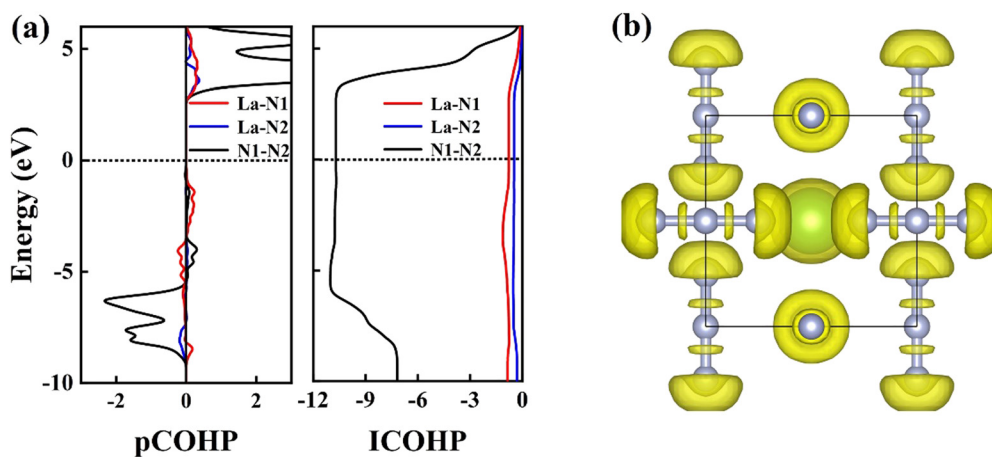


Fig. 4 (a) The projected crystal orbital Hamilton population (pCOHP) and integral crystal orbital Hamilton population (ICOHP) for LaN_9 at 20 GPa. (b) Three-dimensional ELF of LaN_9 at 20 GPa with an isosurface of 0.8.

partially occupied. Specifically, the ICOHP values for N–N, La–N1, and La–N2 interactions are -10.7 , -0.8 , and -0.5 , respectively, signifying a robust N–N bond and relatively weaker La–N interactions.

Subsequently, we conducted electron localization function (ELF) calculations⁶⁸ to further elucidate the bonding properties of LaN_9 . As illustrated in Fig. 4(b), the presence of strong

covalent N–N interaction is substantiated by electron localization observed between nearest-neighbor N atoms. Additionally, the central nitrogens (N2) are sp^2 hybridized, and the terminating nitrogens (N1) of the azide anions exhibit a lone electronic pair. There is an absence of an electron localization state between the La and N atoms, suggesting a weak La–N ionic bond interaction.

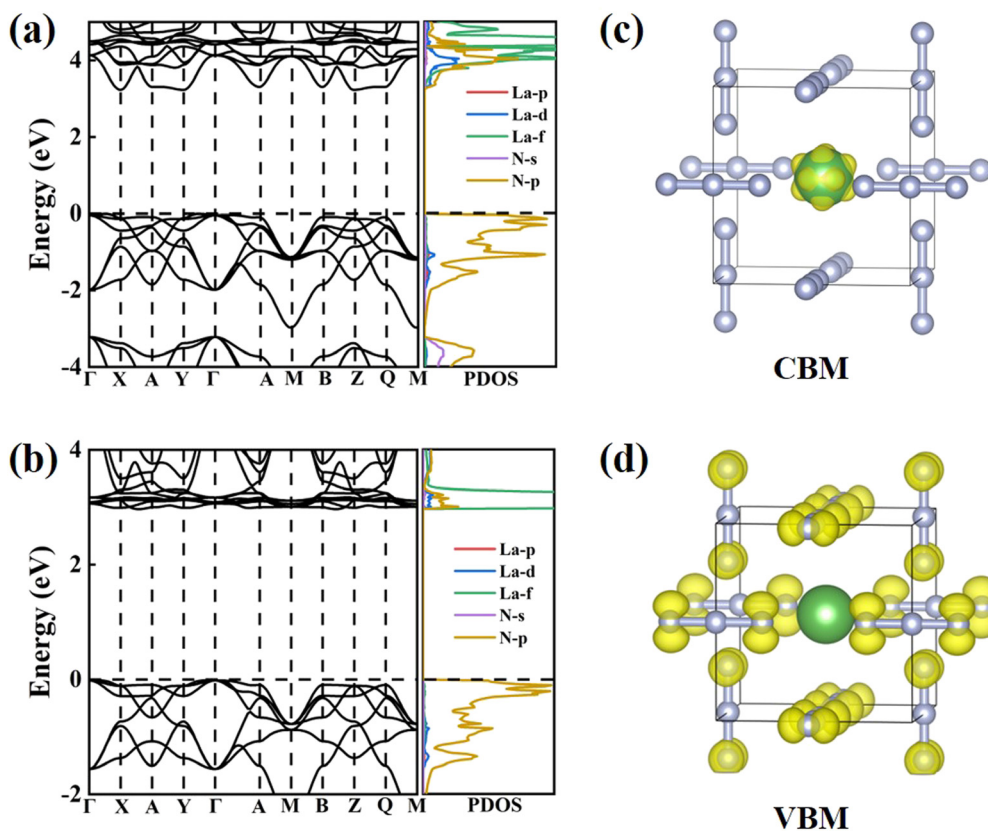


Fig. 5 Band structures and PDOS for LaN_9 at (a) 20 and (b) 0 GPa. The band decomposed charge density for (c) the lowest unoccupied and (d) the highest occupied states of LaN_9 at 0 GPa.

The Bader analysis⁶⁹ suggests that each azide anion receives a certain amount of electrons (0.92 e) from La, providing clear evidence of the ionic relationship between the La atoms and N_3 anions. Based on the preceding analysis, it becomes evident that the exchange of charge between the La and N atoms actively fosters the establishment of La–N ionic bonds and N–N covalent bonds, thereby contributing significantly to the overall structural stability.

The electronic structures and partial density of states (PDOS) of LaN_9 were calculated and are shown in Fig. 5(a) and (b). Our findings indicate that LaN_9 is a wide-gap semiconductor, featuring a band gap of 3.2 eV at 20 GPa. As the pressure decreases to 0 GPa, the band gap reduces to 3.0 eV. The primary contribution to the highest occupied and the lowest unoccupied states of LaN_9 at both 20 and 0 GPa can be predominantly attributed to the N_p and La_f orbitals, respectively. This assertion is supported by the decomposition of the band charge density, as illustrated in Fig. 4(d) and 5(c). The slight hybridization between La and N atoms indicates a weak interaction between La and N atoms.

3.3 Ground-state crystal of Cm LaN_8

As illustrated in Fig. 6(a), regarding LaN_8 , our predictions have unveiled the presence of a novel phase with a lower enthalpy in comparison to the previously reported Cc phase (denoted as Cc LaN_8).⁴⁶ This newly discovered phase adopts a monoclinic

structure with the space group Cm , hence referred to as Cm LaN_8 . It is important to emphasize that despite sharing the same crystallographic system as Cc LaN_8 , Cm LaN_8 manifests a significantly distinct polynitrogen configuration. As depicted in Fig. 6(b), Cm LaN_8 exhibits a one-dimensional infinite N-chain with N–N bond lengths ranging from 1.29 to 1.33 Å. Notably, the structure comprises two interlaced nitrogen chains, distinguished by light gray and wine coloring. As shown in Fig. 6(c), the N–N bonding states are fully occupied, while the antibonding states are partially occupied, leading to covalent bonding between the two neighboring N atoms. The ICOHP value for N–N in Cm LaN_8 is -6.7 (Fig. 6(d)), and its absolute magnitude is smaller than that for N–N in LaN_9 . This suggests the relatively weaker N–N covalent bonding in Cm LaN_8 than that in LaN_9 , attributed to the larger N–N bond length in Cm LaN_8 as compared to LaN_9 .

Phonon dispersion curve calculations and AIMD simulations collectively indicate that Cm LaN_8 exhibits both dynamical and thermal stability under high-pressure conditions (Fig. S2, ESI†). Furthermore, Cm LaN_8 could be retained as a metastable state under atmospheric conditions (Fig. S2, ESI†). At 0 GPa, Cm LaN_8 transforms into the same space group structure. Upon closer examination of the two structures, it became evident that the polymerization mode of nitrogen atoms remained consistent. The primary distinction lay in the manner in which the two N_∞ chains were stacked. The number

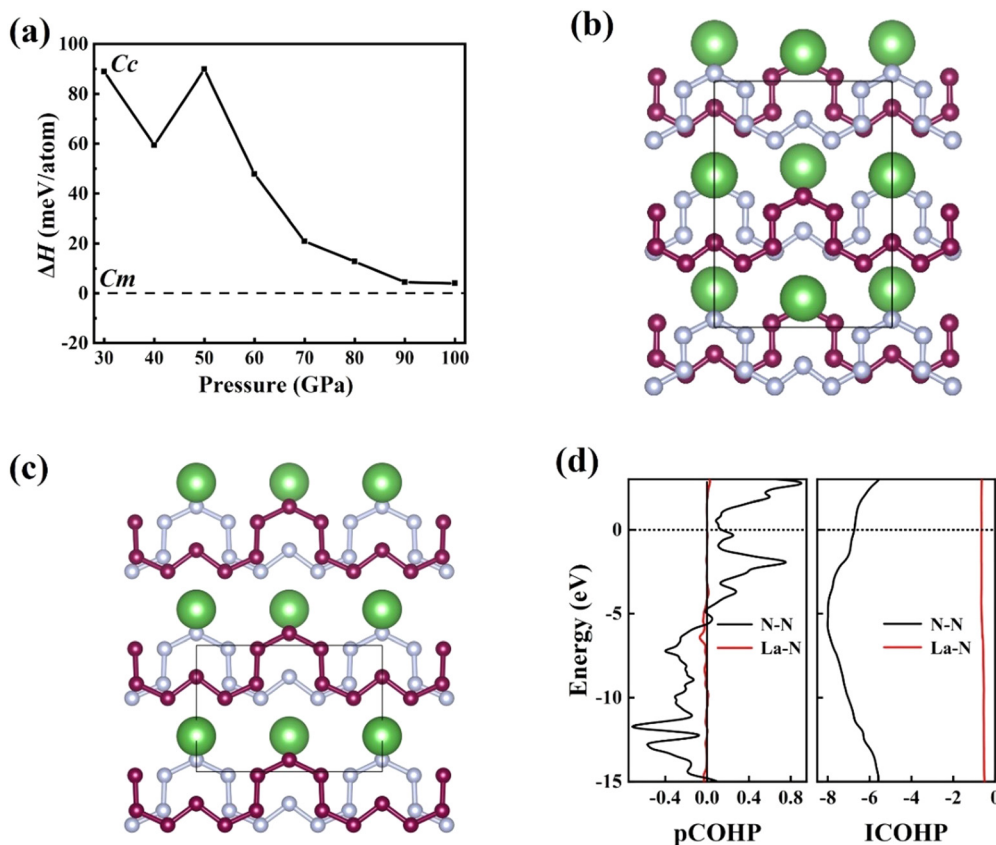


Fig. 6 (a) The enthalpy difference curves of LaN_8 in the Cc phase with respect to the Cm phase. Crystal structure of $cm \text{LaN}_8$ at (b) 60 and (c) 0 GPa. (d) Plots of pCOHP and ICOHP for LaN_8 at 60 GPa.

of atoms in the LaN_8 structure after phase transition is reduced to half of its initial structure. Electronic property calculations indicate the metallic nature of $Cm \text{LaN}_8$, as evidenced by the presence of multiple bands crossing the Fermi level (Fig. S3, ESI[†]).

3.4 Energy density, detonation velocity, and pressure of LaN_9 and LaN_8

The ambient-pressure recoverability of LaN_9 and LaN_8 is deduced from the theoretical calculations. In experimental scenarios, numerous uncertainties emerge during pressure release. Consequently, the assurance of environmental recoverability for LaN_8 and LaN_9 cannot be guaranteed, a consideration that lies beyond the scope of our work. However, we are still optimistic about the recoverability of LaN_9 and LaN_8 . Therefore, we estimate the energy density, detonation velocity, and pressure of two phases as HEDMs at 0 GPa.

Table 1 Calculated mass density (ρ), energy density (E_d), detonation velocity (v_d), and detonation pressure (P_d) of LaN_9 and $cm \text{LaN}_8$

Compounds	ρ (g cm ⁻³)	E_d (kJ g ⁻¹)	v_d (km s ⁻¹)	P_d (kbar)
LaN_9	3.44	0.90	8.02	388.00
$cm \text{LaN}_8$	3.88	1.54	9.62	582.02
TNT ⁵⁹	1.64	4.30	6.90	190
HMX ⁵⁹	1.90	5.70	9.10	393

The dissociation energies of LaN_9 and $Cm \text{LaN}_8$ were calculated by considering the decomposition paths of $\text{LaN}_x \rightarrow \text{LaN}_2 + [(x-2)/2]\text{N}_2$. The results reveal that LaN_9 and $Cm \text{LaN}_8$ can release energies of 0.26 and 0.45 eV per atom, yielding energy densities of 0.90 and 1.54 kJ g⁻¹, respectively. These values are in line with those estimated for SrN_5 (1.09 kJ g⁻¹),²³ FeN_4 (1.37 kJ g⁻¹),⁴⁴ and GdN_6 (1.62 kJ g⁻¹).³⁴ Furthermore, we have investigated their detonation velocity (v_d) and pressure (P_d), and the results are presented in Table 1. It is noteworthy that the calculated v_d for LaN_8 is 9.62 km s⁻¹, placing it on par with HNX and TNT. Additionally, the estimated P_d for LaN_8 reaches a remarkable 582 kbar, significantly exceeding the values associated with HNX and TNT. These exceptional performance characteristics suggest the promising application potential of LaN_9 and $Cm \text{LaN}_8$ in the fields of energetic materials and explosives.

4. Conclusions

In summary, our study involved a comprehensive exploration of the high-pressure phase diagrams within the La-N binary system, utilizing the CALYPSO method in conjunction with first-principles calculations. In addition to the previously reported La-N compounds, we have proposed the existence of four stable high-pressure phases, namely LaN_4 , LaN_9 , LaN_8 , and

LaN₃. Of particular significance, LaN₉ comprises linear N₃ anion units, holding the uniqueness of being the first thermodynamically stable metal nine-nitrogen compound. Our analysis reveals that charge transfer between the La and N atoms plays a crucial role in enhancing structural stability, facilitating the formation of La–N ionic bonds and N–N covalent bonds. Furthermore, we have identified a novel *Cm* phase for LaN₈ characterized by a lower enthalpy in comparison to the previously reported phase. In *Cm* LaN₈, nitrogen atoms are polymerized into infinite N_∞ chains. Both LaN₉ and *Cm* LaN₈ can be quenched under ambient conditions while retaining their initial polynitrogen configuration. LaN₉ and *Cm* LaN₈ exhibit excellent detonation pressures and velocities compared to TNT and HMX. The estimated energy densities of LaN₉ and *Cm* LaN₈ are 0.9 and 1.54 kJ g^{−1}, respectively, comparable to those of most metal nitrides. Results show their potential for application as HEDMs. This research offers a structural prototype for rare earth metal nine-nitrides and valuable insights for the exploration of novel high-energy lanthanide polynitrides.

Conflicts of interest

The authors declare no competing financial or nonfinancial interests.

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